eclipse by comparing observations of starlight that just grazed the sun's surface (these are the light rays most affected by the sun's presence, and only during an eclipse are they visible) with photographs taken when the earth's orbit had placed it between these same stars and the sun, virtually eliminating the sun's gravitational impact on the starlight's trajectory. The comparison revealed a bending angle that, once again, confirmed Einstein's calculations. When the press caught wind of the result, Einstein became a world-renowned celebrity overnight. With general relativity, it's fair to say, Einstein was on a roll.

Yet, despite the mounting successes of general relativity, for years after he first applied his theory to the most immense of all challenges—understanding the entire universe—Einstein absolutely refused to accept the answer that emerged from the mathematics. Before the work of Friedmann and Lemaître discussed in Chapter 8, Einstein, too, had realized that the equations of general relativity showed that the universe could not be static; the fabric of space could stretch or it could shrink, but it could not maintain a fixed size. This suggested that the universe might have had a definite beginning, when the fabric was maximally compressed, and might even have a definite end. Einstein stubbornly balked at this consequence of general relativity, because he and everyone else "knew" that the universe was eternal and, on the largest of scales, fixed and unchanging. Thus, notwithstanding the beauty and the successes of general relativity, Einstein reopened his notebook and sought a modification of the equations that would allow for a universe that conformed to the prevailing prejudice. It didn't take him long. In 1917 he achieved the goal by introducing a new term into the equations of general relativity: the *cosmologi*cal constant.[4]

Einstein's strategy in introducing this modification is not hard to grasp. The gravitational force between any two objects, whether they're baseballs, planets, stars, comets, or what have you, is attractive, and as a result, gravity constantly acts to draw objects toward one another. The gravitational attraction between the earth and a dancer leaping upward causes the dancer to slow down, reach a maximum height, and then head back down. If a choreographer wants a static configuration in which the dancer floats in midair, there would have to be a repulsive force between the dancer and the earth that would precisely balance their gravitational attraction: a static configuration can arise only when there is a perfect